

BIOS 755: Missing Data in Longitudinal Studies II

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Some Simple Approaches

- ▶ We will now discuss some simple methods people use to deal with missing data.
- ▶ These methods usually create a single 'complete' dataset, which is analyzed as if it were the fully observed data.
- ▶ Unless certain, assumptions are true, the answers are biased or invalid.

Complete Case Analysis

- First, we will discuss complete case analysis:

Subject	Y_1	Y_2
1	2	-0.668
2	0	-2.118
3	1	-0.480
4	1	-1.058
5	2	??

Complete Case Analysis

- ▶ Completer case analysis (CCA) deletes all units with incomplete data (in the variables involved) from the analysis.
- ▶ Can be inefficient.
- ▶ Need to be careful comparing models when using CCA
- ▶ Need to make sure you are not changing the size of the data set, as you add/remove explanatory variables with missing observations, or we use the subset of the data with no missing values.
- ▶ When the missing observations are not MCAR or MAR (depends on the model, will discuss below), this analysis may give biased estimates and invalid inferences.
- ▶ The default in statistical packages.

Imputation of the mean

Subject	Y_1	Y_2
1	2	-0.668
2	0	-2.118
3	1	-0.480
4	1	-1.058
5	2	-0.883

- ▶ This approach is clearly inappropriate for categorical variables.
- ▶ It does not lead to proper estimates of measures of association or regression coefficients. Rather, associations tend to be diluted.
- ▶ In addition, variances will be wrongly estimated (typically under estimated) if the imputed values are treated as real. Thus inferences will be wrong too.

Regression mean imputation

- ▶ In this case we use those with complete data to build a model that relates all variables.
- ▶ Then we substitute in the *predicted mean* for each unit with a missing value.
- ▶ In this way we use information from the joint distribution of the variables to make the imputation.
- ▶ In our simple data set we might consider

$$Y_2 = \beta Y_1 + \varepsilon \quad \varepsilon \sim N(0, \sigma^2)$$

- ▶ This is a better idea than mean imputation, but not good in this form.
- ▶ This is the idea behind multiple imputation that we will discuss later on.

Creating an extra category

Subject	Y_1	Y_2
1	2	-0.668
2	0	-2.118
3	? → 3	-0.480
4	1	-1.058
5	? → 3	0.453

- ▶ The impact of this strategy depends on how missing values are divided among the real categories, and how the probability of a value being missing depends on other variables;
- ▶ Very dissimilar classes can be lumped into one group;
- ▶ Sever bias can arise, in any direction, and
- ▶ When used to correct for confounding the completed categorical variable will not do its job properly.

Last Observation Carried Forward (LOCF)

Subject	Y_1	Y_2	Y_3	Y_4	Y_5
1	-0.668	0.223	0.119	0.908	0.171
2	-2.118	-0.941	0.465	? → 0.465	-0.554
3	-0.480	1.387	? → 1.387	? → 1.387	-0.218
4	-1.058	-0.413	-0.243	0.136	-0.106
5	-0.092	-0.377	1.115	0.215	-1.220

- ▶ Using LOCF, once the data set has been completed in this way it is analyzed as if it were fully observed.
- ▶ For full longitudinal data analyses this is clearly disastrous:
 - ▶ Means and covariance structure are seriously distorted.
 - ▶ For single time point analyses the means are still likely to be distorted, measures of precision are wrong and hence inferences are wrong.

Summary of 'ad hoc' methods

- ▶ Unless the proportion missing is so small as to be unlikely to affect inferences, these simple ad-hoc methods should be avoided.
- ▶ They usually conflict with the statistical model that underpins the analysis (however simple and implicit this might be) so they introduce bias.
- ▶ As the assumptions about the reason for the data being missing that they implicitly make are often difficult to describe (e.g. with LOCF), they can make it very hard to know what assumptions are being made in the analysis.
- ▶ **They do not properly reflect statistical uncertainty: data are effectively 'made up' and no subsequent account is taken of this.**

Properties of a good missing-data method

- ▶ Makes use of partial information on incomplete cases, for reduced bias, increased efficiency.
- ▶ Frequency valid (“calibrated”) inferences under plausible model for missing data (e.g. confidence intervals have nominal coverage).
- ▶ **Accounts for the uncertainty in the missing-data.**
- ▶ Favor likelihood based approaches, but GEE's are also good.
 - ▶ Maximum Likelihood (ML) for large samples
 - ▶ Multiple Imputation/Bayes for small samples

MAR methods

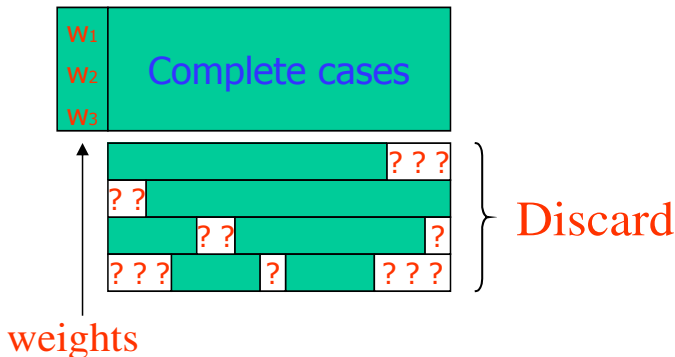
The main approaches to analyzing MAR data are:

- ▶ Weighting methods.
- ▶ Multiple imputation.
- ▶ Direct likelihood.
- ▶ Direct bayesian.

WEIGHTING METHODS

Introduction

- ▶ The complete case (CC) analysis is the default in statistical packages (list-wise deletion).
- ▶ Weighting methods using a CC where the complete cases are weighted.



Weights

- ▶ The weight corresponds to the inverse probability that the data point was collected.

$$w_i = \frac{1}{\Pr(i \text{ completely responds} \mid \mathbf{X})}$$

- ▶ In the analysis subject i represents w_i subjects of the population.
- ▶ That is, if $w_i = 2$ then subject i represents 2 people.
- ▶ People that were not likely to respond will be represented more heavily in the analysis.
- ▶ Since probability of response is not known, we need to estimate it.
 - ▶ This can be done using logistic regression.

Inference from Weighted Data

- ▶ Can use packages for computing estimates and standard errors with weighted data (e.g., weighted GEE) – but often these do not take into account sampling uncertainty in weights.
- ▶ Bootstrapping weighting procedures propagates uncertainty in weights – but weights need to be recalculated on each bootstrap sample.
- ▶ Weighting is a relatively simple device for reducing bias from complete-case analysis.
- ▶ Most useful when the pattern of dropout is monotone.
- ▶ These methods include:
 - ▶ Weighted GEE
 - ▶ Propensity Weighting
 - ▶ Weighted mean

MULTIPLE IMPUTATION

Introduction

- ▶ Create multiple data sets, each with a different set of imputed values
- ▶ Each imputed value includes stochastic variation based on the variability in the posterior distribution of the estimate and the residuals
 - ▶ Thus, the imputed values vary across each of the MI data set
- ▶ MI relies on the MAR assumption; a weaker assumption than MCAR

Actual data set

Subject	<i>PA</i>	<i>Age</i>	<i>SBP</i>	<i>BMI</i>	<i>Loss</i>
1	8	32	120	19	0
2	3	48	145	XX	1
3	4	25	XX	XX	0
⋮	⋮	⋮	⋮	⋮	⋮

- Let $Y_{im} \sim N(\hat{\beta}\mathbf{X}_i, \hat{\sigma}^2)$, which is from

$$Y_i = \beta\mathbf{X}_i + \varepsilon \quad \varepsilon \sim N(0, \sigma^2)$$

is fitted using the complete cases.

- Here Y_i is any variable with missingness (usually not the outcome, but can be).

Imputed data sets

Actual data set

Subject	m=1				
	<i>PA</i>	<i>Age</i>	<i>SBP</i>	<i>BMI</i>	<i>Loss</i>
1	8	32	120	19	0
2	3	48	145	28.947	1
3	4	25	132.54	26.426	0
⋮	⋮	⋮	⋮	⋮	⋮
Subject	m=2				
	<i>PA</i>	<i>Age</i>	<i>SBP</i>	<i>BMI</i>	<i>Loss</i>
1	8	32	120	19	0
2	3	48	145	30.126	1
3	4	25	128.14	22.36	0
⋮	⋮	⋮	⋮	⋮	⋮

Multiple Imputation

- ▶ MI is NOT about finding “the best” value for each missing value.
- ▶ MI is about preserving unbiased estimates of population parameters:
 - ▶ means, variances, covariances, correlations, multivariate regression coefficients
 - ▶ **We need to be right on average** (MAR assumption) while accurately representing our uncertainty about missing values.
- ▶ MI uses one model to create the multiple imputed data sets, then another model to analyze the data.
- ▶ When we analyze the data there are methods to control for the imputation.

Multiple Imputation model

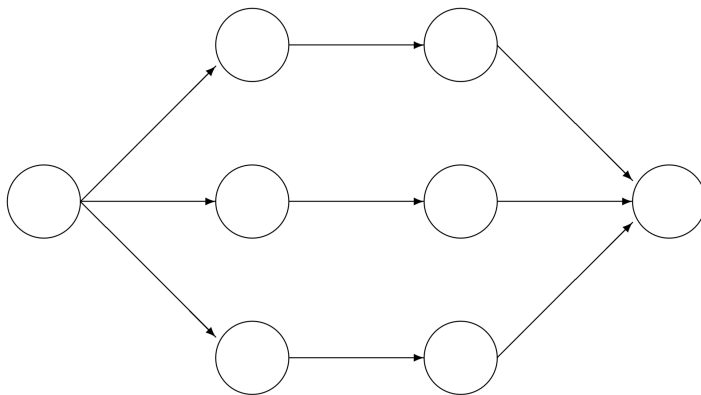
- ▶ Mostly people will use a Multivariate Normal (MVN) MI model for continuous data.
 - ▶ Strongest theoretical justification
 - ▶ Implemented in R, SAS, Stata, and stand alone packages
- ▶ To use this model you need to assume the variables follow a normal distribution.
- ▶ For this reason it is common to transform the variables, perform the MI, then back-transform the variables for analysis of the imputed data sets.

Using Multiple Imputation using fully conditional specification

- ▶ In R, either the Multivariate Imputation by Chained Equations, `mice`, or `Amelia` packages can be used to generate the imputed datasets.
 - ▶ their syntax is very similar.
- ▶ In SAS, there is the fully conditional specification (fcs) model which is similar to `mice`.
- ▶ Benefit of these approaches are that they can be used for binary, count, or multinomial data (really any kind).

Click [here](#) for more information about `mice`.

Using Multiple Imputation



Incomplete data

Imputed data

Analysis results

Pooled result

Summary of Multiple Imputation

- ▶ Use at least 50 data sets.
- ▶ The Imputation model should be at least as comprehensive as the estimation model
 - ▶ The imputation model must include all of the variables & interactions you want to use in the estimation
 - ▶ Include any variables associated with the mechanism of missingness or with the variables to be imputed
- ▶ **DO include the dependent/outcome variable in the imputation model**
- ▶ DO NOT round imputed values (e.g., use a normal model to impute a 0/1 variable)
- ▶ Impute at the lowest level of data possible.
 - ▶ For example, suppose $Y_i = Y_{i1} + Y_{i2} + Y_{i3} + Y_{i4}$, if Y_{i3} is missing then impute Y_{i3} and recalculate Y_i

Likelihood Based Inference

In Likelihood Inference Ignoring the Missing Data Mechanism is valid if

- ▶ Model for Y is correctly specified
- ▶ Data are MAR
- ▶ Fully efficient if “distinctness” condition holds (technical condition on the parameter spaces)
- ▶ This also holds for Bayesian inference methods.
- ▶ See Little and Rubin (2002, chapter 6) for more on this.

In contrast, many ad-hoc methods require the stronger MCAR assumption

Summary methods that can be used in MNAR

- ▶ **Selection model** (Rubin, 1976)

$$\Pr(\mathbf{Y}_{\text{obs}}, \mathbf{R}|\theta) = \Pr(\mathbf{R}|\mathbf{Y}_{\text{obs}}, \theta_T) \Pr(\mathbf{Y}_{\text{obs}}|\theta_Y)$$

model the outcome marginally and appended by a model on the dropout, conditioning on the outcome.

- ▶ **Pattern-mixture model** (Little, 1993)

$$\Pr(\mathbf{Y}_{\text{obs}}, \mathbf{R}|\theta) = \Pr(\mathbf{Y}_{\text{obs}}|\mathbf{R}, \theta_{Y|R}) \Pr(\mathbf{R}|\theta_R)$$

here we condition the outcome on the dropout structure (quality of life).

- ▶ **Shared parameter models** (Wu and Carroll, 1988, Follmann and Wu, 1995)

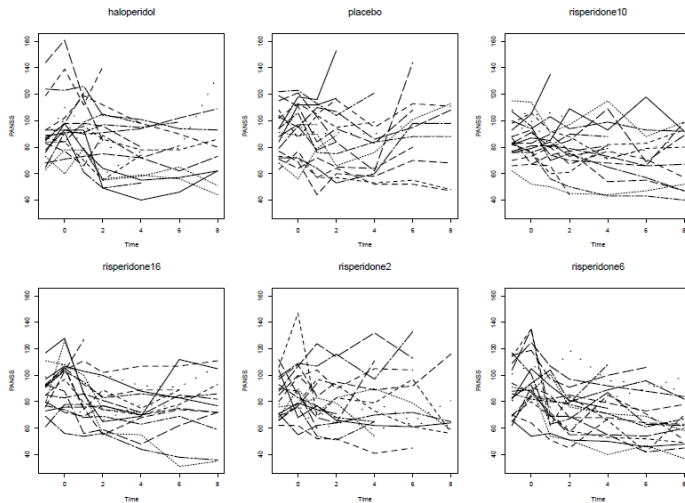
$$\Pr(\mathbf{Y}_{\text{obs}}, \mathbf{R}|\theta) = \int \Pr(\mathbf{R}|b, \theta_{R|b}) \Pr(\mathbf{Y}_{\text{obs}}|b, \theta_{Y|b}) f(b) db$$

can provide useful information on both models.

MNAR Conclusions

- ▶ All the MNAR methods have strong, untestable assumptions most useful in sensitivity analysis.
- ▶ Nonignorable mechanisms can be included in a missing-data analysis, but this is a difficult modeling problem.
- ▶ Often little is known about the missing-data mechanism, and results may be sensitive to formulation.
- ▶ These are a fruitful area of statistical research.

PANSS Example



PANSS Example

	Final Outcome (8wks)		
	est.	(s.e.)	p-value
Haloperidol is reference group			
Risperidone (6mg)	-7.901	(4.148)	0.0580
Placebo	3.962	(4.844)	0.4142
	Longitudinal Analysis		
	est.	(s.e.)	p-value
Haloperidol is reference group			
Risperidone (6mg)	-13.670	(4.256)	0.0014
Placebo	13.085	(4.586)	0.0045

PANSS Example

- ▶ Analysis using available-case data at 8 weeks yields different conclusions from a longitudinal analysis.
- ▶ The result suggests dropout predicts the outcome:

$$E\{Y(8)|\text{completer}\} < E\{Y(8)|\text{drop-out}\}$$

- ▶ Therefore, the data are not MCAR.
- ▶ Q: What is the anticipated impact when analyzing data that are MAR?

General strategies

- ▶ Complete-case (CC) analysis (i.e., discard cases with incomplete data)
 - ▶ CC may be a biased sample
 - ▶ inefficient
 - ▶ weighting of complete cases can be used to correct for bias
- ▶ Imputation
 - ▶ naive imputations (e.g., means, LOCF) can be worse than CC analysis
 - ▶ multiple imputation good (Schafer & Graham, 2002)
- ▶ Analyze as incomplete
 - ▶ different methods have different mechanism assumptions

Incomplete Data Models

- ▶ Distinguish between dependent variable $\mathbf{y} = \mathbf{y}^O$ for $R = 1$ and $\mathbf{y} = \mathbf{y}^M$ for $R = 0$. and independent variables (all observed): $\mathbf{X}$ time, group,
- ▶ GEE assumes special case of MCAR

$$P(\mathbf{R}|\mathbf{y}, \mathbf{X}) = P(\mathbf{R}|\mathbf{X}) \text{ for all } \mathbf{y}$$

$\Rightarrow$ conditional on covariates, R is independent of both $\mathbf{y}^O$ and $\mathbf{y}^M$
(**“covariate-dependent missingness”**)

Incomplete Data Models

- ▶ Likelihood-based methods assume MAR

$$P(\mathbf{R}|\mathbf{y}, \mathbf{X}) = P(\mathbf{R}|\mathbf{X}, \mathbf{y}^O) \text{ for all } \mathbf{y}^M$$

⇒ conditional on covariates and observed values of the dependent variable, $\mathbf{R}$ is independent of $\mathbf{y}^M$ “ignorable non-response” (Laird, 1988)

- ▶ But if baseline covariates are missing, all their $\mathbf{y}$ data are missing from our analyses (whether they were observed or not).
- ▶ In that situation, we'd need to assume that the probability they are missing is independent of the $\mathbf{y}$ data.

Simulation Study

- Data from 5000 subjects were simulated according to:

$$y_{ij} = \beta_0 + \beta_1 T_j + \beta_2 G_i + \beta_3 (G_i \times T_j) + v_{0i} + v_{1i} T_j + \varepsilon_{ij}$$

$T_j = 0, 1, 2, 3, 4$ for five timepoints

$G_i =$ dummy-code (0 or 1) with half in each group

Regression coefficients:

$$\beta_0 = 25, \beta_1 = -1, \beta_2 = 0, \text{ and } \beta_3 = -1$$

$\Rightarrow$ the population means are:

25, 24, 23, 22, and 21 for $Grp = 0$

25, 23, 21, 19, and 17 for $Grp = 1$

Variance parameters:

$$\sigma_{v_0}^2 = 4, \sigma_{v_1}^2 = .25, \sigma_{v_{01}} = -.1 \ (\rho = -.1), \sigma^2 = 4$$

Simulation Study

- The population variance-covariance matrix,

$$V(\mathbf{y}) = \mathbf{Z}\Sigma_v\mathbf{Z}' + \sigma^2\mathbf{I}$$

$$V(\mathbf{y}) = \begin{bmatrix} 8.00 & 3.90 & 3.80 & 3.70 & 3.60 \\ 3.90 & 8.05 & 4.20 & 4.35 & 4.50 \\ 3.80 & 4.20 & 8.60 & 5.00 & 5.40 \\ 3.70 & 4.35 & 5.00 & 9.65 & 6.30 \\ 3.60 & 4.50 & 5.40 & 6.30 & 11.20 \end{bmatrix}$$

Scenarios

- ▶ **Complete data:** no missing data.
- ▶ **50% random missing:** 50% missing data at every timepoint; completely random and unrelated to any variable.
- ▶ **Time related dropout:** dropout rates of 0%, 25%, 50%, 75%, 87.5% for the five timepoints. If a subject was missing at a timepoint, then they were also missing at all later timepoints; these rates indicate the percentage of the original sample that were missing at each of these timepoints.

Scenarios

- ▶ **Group by time related dropout:** dropout rates of
0%, 23%, 46%, 70% and 83% for $G=0$
0%, 27%, 55%, 81% and 91% for $G=1$
⇒ notice that these missing data scenarios are all MCAR (as long as analysis model includes time, G , and G by time)

Results

► MCAR Simulation Results – Mixed effects Model estimates (standard errors)

	β_0	β_1	β_2	β_3	$\sigma_{v_0}^2$	$\sigma_{v_{01}}$	$\sigma_{v_1}^2$	σ^2
simulated value:	25	-1	0	-1	4	-.1	.25	4
complete data	24.969 (.050)	-.994 (.016)	-.001 (.071)	-.986 (.023)	3.918 (.129)	-.057 (.032)	.239 (.014)	3.991 (.046)
50% random missing	24.991 (.063)	-1.024 (.023)	-.087 (.089)	-.933 (.032)	3.811 (.193)	-.020 (.056)	.199 (.025)	4.070 (.083)
Time-related dropout	24.989 (.053)	-.968 (.028)	.019 (.075)	-1.021 (.040)	3.853 (.150)	-.062 (.060)	.229 (.032)	4.000 (.074)
Group by Time related dropout	24.991 (.053)	-.977 (.026)	.041 (.075)	-1.014 (.041)	3.872 (.150)	-.048 (.059)	.234 (.031)	3.994 (.073)

MAR and MNAR Scenarios

- ▶ **MAR(a):** if the value of the dependent variable was lower than 23, then the subject dropped out at the next timepoint (i.e., they were missing at the next and all subsequent timepoints).
- ▶ **MAR(b):** the MAR specification was different for the two groups. For $\text{Grp} = 1$, if the dependent variable was lower than 23, then the subject dropped out at the next timepoint, however for $\text{Grp} = 0$, if the dependent variable was greater than 25.5 then the subject dropped out at the next timepoint.
- ▶ **MNAR:** after the first timepoint, if the value of the dependent variable was lower than 21.5, then the subject was missing at that timepoint and all subsequent timepoints.

MAR and MNAR Simulation Results - Estimates (standard errors)

	β_0	β_1	β_2	β_3	$\sigma_{v_0}^2$	$\sigma_{v_{01}}$	$\sigma_{v_1}^2$	σ^2
simulated value:	25	-1	0	-1	4	-.1	.25	4
<u>MAR(a)</u>								
MRM	24.996 (.053)	-1.039 (.025)	-.010 (.075)	-.969 (.041)	3.981 (.158)	-.064 (.065)	.233 (.032)	3.873 (.078)
GEE1	25.281 (.058)	-1.164 (.037)	.019 (.097)	-1.001 (.085)				
<u>MAR(b)</u>								
MRM	24.999 (.053)	-1.003 (.022)	-.016 (.075)	-1.004 (.039)	4.050 (.154)	-.082 (.064)	.229 (.027)	3.812 (.073)
GEE1	24.635 (.055)	-.714 (.030)	.634 (.097)	-1.532 (.090)				
<u>MNAR</u>								
MRM	24.956 (.049)	-.233 (.020)	.027 (.070)	-.552 (.035)	3.856 (.131)	-.943 (.051)	.319 (.025)	3.020 (.053)
GEE1	25.051 (.049)	-.386 (.020)	.016 (.071)	-.583 (.034)				

Misspecification of Variance-Covariance

- Re-analysis of the simulated MAR-generated data, however using only a random-intercepts model:

$$y_{ij} = \beta_0 + \beta_1 \text{Time}_j + \beta_2 \text{Grp}_i + \beta_3(\text{Grp}_i \times \text{Time}_j) + v_{0i} + e_{ij}$$

- This is a misspecified model for the variance-covariance structure because the random slope term was omitted from the analysis.

Mis-specified MAR Simulation Results - Estimates (standard errors)

	β_0	β_1	β_2	β_3	$\sigma_{v_0}^2$	$\sigma_{v_{01}}$	$\sigma_{v_1}^2$	σ^2
simulated value:	25	-1	0	-1	4	-.1	.25	4
MRM with MAR(a)	24.938 (.053)	-.891 (.021)	-.009 (.075)	-.949 (.036)	3.722 (.136)			4.329 (.072)
MRM with MAR(b)	24.998 (.053)	-1.048 (.018)	-.069 (.076)	-.805 (.035)	3.880 (.138)			4.288 (.070)

Results

- ▶ These random-intercepts analyses yield biased results, in particular for the time-related parameters β_1 and β_3 .
- ▶ Performing any full-likelihood analysis, even with missing data following an MAR mechanism, does not guarantee that the correct results will be obtained (need to have the mean structure and variance-covariance structure of $\mathbf{y}$ correctly modeled).

Testing MCAR

- ▶ If MCAR, then either GEE or Mixed Effects models are fine (provided that the covariate matrix $\mathbf{X}_i$ includes predictors of missingness).
- ▶ If MAR, then GEE does not perform well, whereas Mixed Effects models analysis is acceptable (as long as the mean and variance-covariance structures are correctly modeled).
- ▶ It is useful to determine whether MCAR is acceptable or not.
- ▶ Distinction between MCAR and MAR is that missingness cannot depend on observed values of the dependent variable, $\mathbf{y}_i^O$, in the former, but can in the latter.
- ▶ Tests of MCAR are based on analyses involving $\mathbf{y}_i^O$.

Testing MCAR in 2-timepoint study

- ▶ Suppose all subjects have data at time 1, but some are missing at time 2
- ▶ Define $D_i = 0$ for subjects with data at both timepoints, $D_i = 1$ for subjects with data at first timepoint only
- ▶ Compare y_1 between these two groups ($D_i = 0$ vs. $D_i = 1$); for MCAR y_1 data should not differ between the groups.
- ▶ Logistic regression model

$$\log \left\{ \frac{P(D_i = 1)}{1 - P(D_i = 1)} \right\} = \alpha_0 + \alpha_1 y_{i1} + \alpha_2 \mathbf{x}_i + \alpha_3 (y_{i1} \times \mathbf{x}_i)$$

- ▶ α_1 and α_3 are vectors of regression coefficients for $\mathbf{x}_i$ and their interactions with $y_{i1} \Rightarrow$ MCAR dictates that $\alpha_1 = \alpha_3 = 0$.

Testing MAR vs MNAR

- ▶ The only way to test the MAR assumption (i.e., that $\mathbf{R}$ is independent of $\mathbf{y}^M$ given $\mathbf{y}^O$ and $\mathbf{X}_i$) is to have some measure on the missing data.
- ▶ A follow-up with phone calls to a group of non-respondents can accomplish this.
- ▶ Mostly, we will not have any data.
- ▶ The only quantitative solution, in this case, is to analyze with a MAR model and an MNAR model (i.e., selection, pattern-mixture, or shared parameter).
- ▶ The most common way to “test” the MAR assumption is to **use your scientific knowledge of the data and the field.**